

THE TREASURE MAP GRID

Mastering Coordinates via Over-and-Up Footsteps

GRADE 3 • EXPLORER MANUAL



Total Duration: 30 Minutes

Scope: 5x5 Quadrant 1 Grid | Core Rule: Over-then-Up

The Story Hook: The Ancient Fort Secret Map

An Exciting Discovery!

Young **Vihaan** and his sister **Meher** are visiting their grandfather at a historic heritage property near the famous **Amber Fort in Jaipur**. The ancient stones around them are weathered and full of mysterious history.

While exploring an old wooden chest in the dusty attic, they uncover a small, rolled-up parchment map detailing the fort's inner courtyard layout!

Cracking the Code

The map is overlaid with a clean grid of **5x5 square stone tracks**. Look at what they see:

"Look, Meher! A shiny gold star is stamped on one specific stone square! But look at the bottom of the page—it just says the numbers 3 and 4 next to each other. How do we use these to find the hidden treasure?"

— Vihaan

Meher observes that counting blindly leads them to completely separate corners. She prompts Vihaan to think like a real **Map Cartographer**, inviting him to join Grandfather in cracking the secret direction rules of the grid lines!

Meeting the Grid Baselines

A map grid relies on two primary baseline tracks that meet at the bottom-left corner. These tracks act as our guide metrics for measuring distances across the layout:

 Map Line Name	 Everyday Tracker	 Physical Step Direction
X-Axis	The Flat Floor Line	Measures horizontal footsteps traveling sideways (Over).
Y-Axis	The Standing Wall Line	Measures vertical footsteps climbing straight up (Up).

Home Base: The Central Starting Flag

Before any player can run or log a step across our 5x5 matrix arena, they must plant their feet exactly at the starting corner flag position.

Playground Link!

This starting spot functions exactly like the central boundary line or the starting flag box in a traditional game of **Kho-Kho** or **Kabaddi**. It is the steady anchor where every journey begins!

In math, we call this origin point **Home Base**, and it carries the baseline address of **(0, 0)**.

Resetting the Tracker: Whenever a new mapping search task begins, our tracking avatar must automatically return to this origin point box, resetting its step markers back to zero before executing a new set of instructions.

The Golden Rule: Over-Then-Up!

To find any item or coordinate point on a map grid, you must follow one strict, golden sequence rule layout without exception:

☀️ THE WALK-BEFORE-CLIMB RULE ☀️

ALWAYS WALK OVER FIRST,
THEN CLIMB UP SECOND!

Finding Your Seat at PVR Cinemas

Think about finding your assigned seat number before a movie starts at a local cinema complex like **PVR Cinemas**. You don't jump randomly across rows!

First, you walk sideways along the flat main corridor row to find the matching column line (**Over**). Once you reach the right section, you turn and climb up the stairs to your exact seat row number (**Up**). This unvarying habit mirrors how coordinate pairs operate!

Visual Modeling: Monkey's Banana Hunt

A cute cartoon monkey character starts at home base **(0, 0)**. A sweet yellow banana is hidden on the grid layout, carrying the coordinate address of **(3, 2)**.

Tracking the Monkey's Path Steps:

1. The monkey starts at **Home Base (0,0)**.
2. He takes exactly **3 discrete hops OVER** along the bottom floor line.
3. From that position, he turns and climbs exactly **2 discrete hops UP** the wall line, landing perfectly on the banana!

The console logs the tracking coordinates as a team: **(3, 2)**. The first value shows the horizontal movement (**Over**), while the second value shows the vertical height (**Up**).

The Target Address: Reading Brackets

The Two-Number Address Team

To record a grid spot location on our explorer map, we write the two steps together inside parentheses, separated by a clean comma separator marker: **(3, 4)**.

Decoding the Map Coordinates:

When an explorer looks at this bracket team, they read it from left to right as a simple sequence of steps:

- The **left number (3)** commands them to take 3 footsteps **Over** along the floor line.
- The **right number (4)** commands them to take 4 footsteps **Up** the wall line, guiding them directly to their destination!

Agricultural Layouts: The Plant Nursery Grid

Inside a local plant nursery (*Nursery Talab*), hundreds of delicate flower saplings are arranged in a neat, orderly grid matrix to make watering and tracking simple. By using these coordinate targets, the nursery team can pinpoint any individual plant right away:

 Target Plant	 Nursery Address	 Footstep Navigation Path
Jasmine Sapling	(4, 3)	Walk over to Lane 4 , then count 3 plants up the row.
Marigold Bloom	(1, 5)	Walk over to Lane 1 , then count 5 plants up the row.

Laboratory Phase: Secret Hangar Hiding Game

Now, step into the role of map experts! You take command of a virtual archaeological site explorer rover waiting at **Home Base (0, 0)**, inputting coordinate pairs to steer the vehicle toward hidden targets on a 5x5 grid track.

ON-SCREEN INTERFACE SIMULATION:

Active Mission Workspace: A grid coordinate selection card appears on screen. Adjust the numerical selectors to input targets, then click the green launch button to watch the rover trace its path along the lines!

Hangar Run 1: Uncovering the Ancient Coin

Targeting Coordinate Address: **(2, 4)**

Rover Track Output: Over 2 Hops → Up 4 Hops

The vehicle traces the floor line for 2 units, then climbs up 4 units, successfully uncovering an **ancient coin!**

Hangar Run 2: The Reversed Order Misconception

Imagine a student is trying to reach a target at (2, 4), but they reverse the steps, taking 4 footsteps up the wall line first and then moving 2 footsteps over. They land at **(4, 2)** instead!

System Feedback Warning:

*"Oops! Your path went Up first and Over second, causing the rover to go off course and land on a **prickly cactus patch!** Remember our golden rule: always walk across the floor line before climbing up the wall."*

Tracking Ground-Level & Data Synthesis

🎯 Hangar Run 3: Tracking Ground-Level Coordinates

For our final exploration run, we test a unique coordinate target that contains a zero value:
Targeting Coordinate Address (5, 0).



Rover Track Output: Moves Over 5 Hops → Stays on Ground Zero

Because the vertical height value is exactly zero, the rover doesn't climb up the wall line at all. It finishes its path right on the bottom floor track! Zero means staying grounded on the baseline.



The Explorer's Telemetry Guide

Comparing our three exploration runs side by side makes the directional path patterns clear and easy to follow:

Code Address	Horizontal Floor Steps (Over)	Vertical Wall Steps (Up)	🎯 Destination Outcome
(2, 4)	2 Hops	4 Hops	💎 Safely uncovers the ancient coin .
(4, 2) - <i>Reversed</i>	4 Hops (Wrong Order)	2 Hops	🌵 Goes off course and lands on the prickly cactus patch .
(5, 0)	5 Hops	0 Hops	⚓ Stays grounded directly on the floor baseline.

Concept Review Challenges

Let's confirm that you've mastered our core navigation rules. Select the correct option below!

? Challenge 1: Spotting the First Step

When you read a coordinate address pair like **(1, 3)** on an explorer map, which directional track must your feet follow first from home base?

☒ A) The Flat Floor Line (Over)

B) The Standing Wall Line (Up)

C) Climb the Ladder First

D) Leap Diagonally Across Rows

Why it's correct: Our golden mapping rule commands us to always move horizontal footsteps along the flat floor line first before turning to climb up the wall track.

? Challenge 2: Decoding Map Targets

An ancient gold star is hidden on our map grid. To reach it, an explorer must take exactly **3 footsteps Over** along the floor line and **4 footsteps Up** the wall line. What is its address code team?

A) (4, 3)

☒ B) (3, 4)

C) (3, 0)

Why it's correct: Because the floor steps are written on the left side and the wall steps on the right, moving over 3 and up 4 maps perfectly to the address code team (3, 4).

The Floor Tile Treasure Hunt Quest!

Scientific discovery and math exploration don't have to end in the classroom! You can continue your mapping journey at home with a fun playground challenge.



YOUR HOME MAPPING MISSION:

1. Find a tiled section of floor at home (like in your kitchen or entryway) to act as your map.
2. Pick a specific corner square to be your **Home Base (0, 0)**.
3. Hide a small toy on one of the floor tiles within your grid space.
4. Count the path steps from home base, measuring how many tiles you move **Over** and how many you move **Up** to reach the toy.
5. Write down the address code team on a card and challenge a family member to decode it and find the hidden treasure!



Documenting Your Home Discoveries: Sketch a quick diagram of your tile map grid in a notebook, coloring in your home base and target location squares! Sharing your findings with family is a fantastic way to show off your new cartography skills.



Congratulations, Map Cartographers!

You have successfully completed your very first steps into coordinate geometry, unlocking the path sequences that turn simple grid lines into powerful treasure maps!



MASTER CARTOGRAPHER MILESTONE ACHIEVED

